Executive Summary

Care Transition Efficiency & Placement Outcome Analytics

The **Care Transition Efficiency & Placement Outcome Analytics** project evaluates the movement of unaccompanied children through a multi-stage care pipeline represented as **CBP Custody → HHS Care → Sponsor Placement**. The analysis was designed to assess transition efficiency, discharge performance, potential accumulation pressure, temporal patterns, and stability of placement outcomes using reporting-level operational data.

The dataset contains **720 valid reporting observations** covering approximately **January 2023 to December 2025**. The analysis distinguishes between **stock measures**—children recorded in CBP custody and HHS care at a reporting point—and **flow measures**, including apprehensions, transfers, and discharges. This distinction is important because the dataset represents aggregate system-level activity rather than individual child-level journeys.

Key Findings

The analysis identified substantial variation in the size and activity of the care pipeline over the study period. HHS care levels were consistently much larger than CBP custody levels. HHS care increased from approximately **6,566 children in January 2023** in the sample observations and reached levels above **11,000** during late 2023 and early 2024. During 2024, HHS care levels gradually declined with considerable fluctuation, followed by a pronounced reduction in early 2025. CBP custody remained comparatively small, although periods of increased custody were visible during late 2023 and early 2024.

Three primary transition efficiency metrics were calculated. The **Transfer Efficiency Ratio** was approximately **0.75 (75.03%)**, calculated as transfers divided by CBP custody. The **Discharge Effectiveness** was approximately **0.03 (2.86%)**, calculated as discharges divided by HHS care. The **Pipeline Throughput Rate** was **1.85 (185.42%)**, calculated as total exits divided by total entries. Because these metrics combine flow and stock measures or aggregate observations, they should be interpreted as **operational indicators rather than individual-level transition probabilities**.

Backlog and delay analysis compared **transfers into HHS care with successful exits through discharge**. The sample analysis showed several periods where successful exits exceeded incoming transfers. The cumulative flow-balance analysis remained negative, reaching approximately **−34,000** around early 2024 before stabilizing at lower negative levels. This indicates that cumulative recorded exits exceeded cumulative recorded transfers across the available observations and does not show a sustained inflow-driven accumulation pattern under the defined metric. However, the cumulative measure should not be interpreted as a literal count of unresolved cases because the dataset contains gaps between reporting dates.

Temporal analysis found relatively similar transfer efficiency between weekdays and weekends. Average transfer efficiency was approximately **68.8% for weekdays** and **70.0% for weekends**. Average discharge effectiveness was approximately **2.27% on weekdays** and **2.84% on weekends**. The relatively small differences indicate broadly similar operational ratios across the two day types in the available reporting observations.

A major finding emerged from the **placement outcome stability analysis**. Across 720 observations, discharge volume had a mean of **173.41 children**, standard deviation of **125.70**, median of **181**, minimum of **0**, and maximum of **505**. The high standard deviation relative to the mean demonstrates considerable variability in discharge activity. Discharge volumes were relatively high and highly variable during 2023, gradually declined during 2024, and then experienced a **sharp and persistent reduction around early 2025**.

The stagnation analysis further identified multiple periods of comparatively low discharge activity during 2025. The longest identified low-activity period extended from **August 24, 2025 to December 21, 2025**, covering **82 reporting observations**, with an average of approximately **10.35 discharges per observation**. Another substantial period occurred from **March 12 to June 4, 2025**, covering **59 observations**, with an average of only **9.22 discharges per observation**. These periods represent important signals for operational investigation, although low discharge activity alone does not establish the existence or cause of a backlog.

Analytical Contribution

From a data-analytics perspective, the project transforms the dataset from a simple **care-capacity monitoring dataset into a process-efficiency framework**. Rather than examining CBP custody, HHS care, transfers, and discharges independently, the analysis connects these measures to evaluate the movement and balance of the overall pipeline.

The analytical framework includes:

- **Care Pipeline Modeling** – CBP Custody → HHS Care → Sponsor Placement
- **Transition Efficiency Metrics** – transfer efficiency and discharge effectiveness
- **Pipeline Throughput Analysis** – aggregate exits relative to aggregate entries
- **Backlog and Delay Detection** – inflow versus successful exits and cumulative flow balance
- **Temporal Analysis** – weekday/weekend patterns and monthly placement trends
- **Stagnation Detection** – identification of prolonged low-discharge periods
- **Outcome Stability Analysis** – variability, rolling performance, and potential sudden drops

Business and Operational Implications

The findings highlight the importance of monitoring the care pipeline through a combination of **stock levels, flow volumes, efficiency ratios, and sustained trends**. In particular, the substantial reduction in discharge activity during 2025 warrants further examination alongside HHS care levels and transfer volumes to determine whether the change reflects lower overall pipeline activity, changes in inflow, or other operational factors.

The analysis supports the use of **threshold-based early-warning indicators**, regular monitoring of inflow–exit imbalances, and rolling performance measures. A dashboard implementation can provide decision-makers with the ability to select reporting periods, monitor efficiency ratios, identify potential bottlenecks, and investigate periods of unusually low placement activity.

Important Data Limitation

The dataset is **aggregate and reporting-level rather than individual-level**, and reporting dates are not completely continuous. Consequently, the analysis measures system-level patterns and flow balances rather than individual transition times or individual reunification outcomes. In addition, the dataset does not contain a separate variable explicitly measuring individual **reunification success**; therefore, discharge is used as the available indicator of successful exit/sponsor placement in accordance with the project framework.

Overall Conclusion

The analysis demonstrates that the most significant change in the UAC care pipeline occurred during 2025, when HHS care levels and discharge activity declined sharply and prolonged low-discharge periods emerged. At the same time, the transfer and discharge flow analysis does not show sustained inflow-driven accumulation under the defined net-flow measure. The dataset therefore points to a major change in overall pipeline activity rather than providing, by itself, evidence of a persistent transfer-driven backlog. The combination of a mean discharge volume of 173.41, a standard deviation of 125.70, a maximum of 505, and prolonged 2025 periods averaging only 9.22–15.17 discharges provides a strong quantitative basis for further investigation of the change in placement activity.